ALEXANDER JAMES WHITMORE

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PROFESSIONAL SUMMARY

Research Engineer with 12 years of experience developing machine learning systems and data infrastructure at scale. Specialised in experimental design, prototype development, and bridging the gap between academic research and production systems. Based in Manchester, I've worked across financial services, healthcare analytics, and cloud platforms. Known for building robust data pipelines and mentoring junior engineers through complex technical problems.

WORK EXPERIENCE

Senior Research Engineer

CloudPeak Technologies Ltd

Manchester, United Kingdom

https://cloudpeaktech.io

January 2022 - Present

Designed and implemented large-scale machine learning inference systems processing 2 billion records daily using LangChain frameworks for natural language understanding tasks. Led a team of four engineers to develop automated testing infrastructure that reduced production incidents by 67 percent. Architected a novel feature extraction pipeline in Python and Spark that improved model accuracy by 8.3 percent across 47 downstream applications. Collaborated with research team to translate papers from NeurIPS into production-ready prototypes within 8-week cycles.

Staff Research Engineer

Barclays Bank PLC

London, United Kingdom

https://www.barclays.com

September 2020 - December 2021

Developed end-to-end machine learning systems for fraud detection affecting 8 million customer accounts across multiple geographies. Implemented GitHub Actions CI/CD pipelines for automated model validation and deployment, reducing release cycles from 2 weeks to 3 days. Mentored six junior researchers in experimental methodology and paper writing, resulting in two published works in IEEE journals. Established best practices for reproducible research by creating internal documentation frameworks and standardised testing protocols.

Principal Research Engineer

Synapse Data Ltd

Manchester, United Kingdom

https://synapsedatagroup.net

June 2012 - August 2020

Led research initiatives in biomedical data analytics, focusing on patient outcome prediction models across NHS hospital networks. Engineered scalable data pipelines using Python, Scala, and SQL to process 15 terabytes of heterogeneous clinical data monthly. Pioneered the adoption of the GPT-4 API for automated literature review and hypothesis generation starting in 2018. Supervised three PhD-level researchers and published eight peer-reviewed

papers in medical informatics venues, with a combined 340 citations.

EDUCATION

Doctor of Philosophy (PhD) in Computer Science

University of Manchester

Manchester, United Kingdom

2008 - 2012

Thesis: Scalable Bayesian Methods for Time Series Anomaly Detection

Bachelor of Science (BSc) in Mathematics and Computer Science

University of Leeds

Leeds, United Kingdom

2004 - 2008

TECHNICAL SKILLS

Programming Languages: Python, Scala, Java, SQL, R, Bash

Data Technologies: Apache Spark, Airflow, Kafka, PostgreSQL, Snowflake, DuckDB

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Bayesian inference, statistical testing

Platforms and Tools: AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, Git, Jupyter

Research Methods: Experimental design, hypothesis testing, peer review, literature synthesis, technical writing

PROFESSIONAL MEMBERSHIPS

Fellow of the British Computer Society

Member of the IEEE Computer Society

Member of the Association for the Advancement of Artificial Intelligence